

Born from causality, Tsirelson from steering, and the amplitude seam (DRAFT v2)

Born from causality, Tsirelson from steering, and the amplitude seam:

an internal reconstruction inside a geometric hidden-connection model

Cycle 3 draft v1. Author: A. Oktyabrev (aoktyabrev@gmail.com). Status: assembly draft (~11 pp + battery appendix). All numbers cite raw as file (commit). Claim-discipline per program standard: theorem only with written proof + battery + review; conjecture otherwise.

Abstract

Within a geometric hidden-connection (“ribbon”) model we close, and then delimit, an internal reconstruction of quantum correlations. (i) Where a mechanism carries **steering** — conditional states with internally generated diversity $D(\chi) > 0$ — the no-signaling prohibition forces the Born exponent $p=2$ uniquely; the derivation is internal (the steering ensembles are produced by a shifted source measure, not imported), executable, and robust under five deformations of the measure. (ii) Mechanisms that resolve outcomes **locally** are capped by Bell’s theorem at $S=2$, with a **1/3-cosine point** ($E=\cos\theta/3$) as their isotropic-cosine ceiling, anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it. (iii) The **amplitude seam** between that ceiling and the quantum value $S=2\sqrt{2}$ is not crossed by decorating local mechanisms — three registered attempts fail on a trilemma of horns (no steering / no amplitude / superluminal telegraph) — but only by **joint, nonlocal resolution**, which by our foliation results demands a shared, statistically invisible precedence structure — a preferred foliation being one realization, a frameless shared order-coin another (class-M theorem). The bridge statement is an exact division of labor, not a unification.

1. Introduction — the import and the question

The program’s canonical text acknowledged an **import**: partially entangled ensembles were introduced by hand (“within this ansatz... we make no claim of a general derivation of the Born rule”, FINAL_v1:284-290). Cycle 3 asks whether the steering that underlies the Born rule can be **generated internally** by the model’s own

geometry, and — once it is — what exactly stands between such a mechanism and full quantum correlations. The answer assembles into four registered statements (RULE, CEILING, SEAM, FRAME), each with kill-criteria and public commits.

2. Model

Source / steering (F2). A shared connection orientation λ on S^2 is drawn from a shifted measure $\mu_\chi \propto 1 + \chi \lambda_z$. A measurement setting a partitions λ by $s = \text{sign}(\lambda \cdot a)$; the conditional ensemble's **raw** (sub-unit) mean vector m_s is the steered state; readout probability $f_p(c) = |1 + c|^{\{p/2\}/(1+c)} \{p/2\} + |1 - c|^{\{p/2\}} \}$, with f_∞ the step $1\{c > 0\}$. **Lesson (record):** normalizing the conditional axis broke total probability and produced $\Delta(p=2) > 0$ — the “ $\Delta(2) > 0$ = bug” criterion caught it; the raw sub-unit Bloch vector is correct (battery_c3b, 968aaa5).

Joint / chord layer. A postulated pair law $P(s, t|a, b) = (1 - st \cos \theta)/4$ ($E = -\cos \theta$, $S_{\max} = 2\sqrt{2}$), and its weighted generalization $|s\sqrt{w_a} a - t\sqrt{w_b} b|^p$, used where the mechanism resolves the pair jointly rather than by local readout of λ .

3. RULE — Born from no-signaling, internally (C3-B)

Steering is generated, not imported. With μ_χ , $\chi > 0$ yields steering diversity $D(\chi)$: 0.03→0.08 as χ grows (C3B_hardening.json, 968aaa5) — remote ensembles with **equal means, distinct higher moments**.

Selection lemma (executable). For the readout as a random variable over the conditioning, no-signaling $\Leftrightarrow$ the readout is affine in the projection (Jensen gap of an affine map $\equiv 0$). Affinity $\cap$ oddness ($f(-c) = 1 - f(c)$) $\cap$ normalization = the single rule $f_2 = (1 + c)/2 \Rightarrow p=2$ **uniquely**. The algebraic zero at $p=2$ is exact (addendum2, ccf5af6); the non-affine tail is strictly positive — scan $\Delta(4)=0.042$, $\Delta(6)=0.105$, $\Delta(10)=0.217$, $\Delta(\infty)=0.375$, monotone, analytic $\leftrightarrow$ numeric $< 2\sigma$ (C3L_L3.json, 9567ddb).

Hardened to theorem. T1 (sequential/order, B7): repeatability 1.0000, order commute, no-signaling both orders, post-measurement null-set $\{2\}$. T2 (anti-circularity): **five** measure deformations (μ_κ , $\kappa \in \{0.5, 2, 3\}$; two cubic), all $D > 0$, analytic null-set $\{2\}$, $|\text{analytic} - \text{numeric}| < 2\sigma$. T3 (F1 direct law) is a boundary (postulated, not steering-collapse). Verdict: **internal derivation of the Born rule from no-signaling within the ribbon's steering class** (C3B_hardening.json, c152c98). One **false stop** en route — a quadrature-tolerance gate absent from preregistration — was caught, documented, and corrected without loosening any registered criterion (addendum2, ccf5af6; §7).

Mechanical realization (C3-B-mech, form-free). The steering premise is realized mechanically and form-free: an annealed, field-magnetized and fast-frozen elastic source supplies a polar-biased orientation measure that fits no registered closed form; no closed form is needed. The sign-readout conditionals of that measure carry steering diversity in the third and fifth moments, matching the Jensen analytics computed directly on the empirical ensembles, with the unbiased ($h=0$) control showing null effects throughout; on ensembles produced entirely by the elastic dynamics, no-signaling selects $p=2$. The selection theorem's premise thus rests on measured moments, not on an assumed measure — a weaker assumption and hence a stronger claim (C3Bmech_M1M2.json, 2080bbe; $M1 \checkmark M2 \checkmark$ in form-free mode). That the source measure admits no closed form is kin to the C2-TM

wall (the relaxational amplitude $A(k, f)$ likewise has no closed form): an observation — elasticity does not favor closed forms — not a theorem.

4. CEILING — the 1/3-cosine point

Local resolution (hidden λ + local linear readout) yields $E(a, b) = E_{\lambda}[(\lambda \cdot a)(\lambda \cdot b)] = a^T (\frac{1}{3} I) b = \cos \theta / 3$ (χ -corrections odd $\Rightarrow$ vanish), $S = 2\sqrt{2}/3 \approx 0.943$ — the isotropic-cosine ceiling of the local layer. The exact convergence is analytic (lever rule, exactly 1/3) and frame-local ($E(0) = 0.333$; C3L_L2.json, 49e8c1b); dynamical isotropization lands NEAR the point: $\rho_{\cos} = 0.311 \pm 0.010$ (cosine-fit, the form-matched estimator; raw C2ISO_analysis.json: cells/kf4.0/rho_cos = 0.3113, a043f8f), $\rho_{\text{model-free}} = 0.336$ (same file) — bracketing 1/3 within $\sim 2\sigma$. Estimators named per program rule. Anchored analytically and reached by two mechanisms, with ISO-DYN bracketing it $\Rightarrow$ a structural boundary, the lower bank of the seam.

5. SEAM — the trilemma of horns (C3-S)

Can an internal model reach $D > 0$ and $S = 2\sqrt{2}$ together? Registered attempts, ansätze fixed before batteries: - **S-F1** (weighted chord, $\alpha = \sqrt{1+\chi}$, $\beta = \sqrt{1-\chi}$): $S(\chi) = 2\sqrt{2} \cdot \sqrt{1-\chi^2}$ — reaches $2\sqrt{2}$ only at $\chi = 0$ (symmetry), but is postulated (no continuous measure) $\Rightarrow D_{\text{joint}} \text{ N/A}$. **Horn: no internal steering**. - **S-F2** (shifted source $\times$ chord/product mixture): $S(\chi) = \sqrt{2} \cdot (1 + \chi/3) \leq 1.886$, and $P(s|a, \lambda) = \frac{1}{2} \Rightarrow$ Alice does not update $\lambda \Rightarrow D_{\text{joint}} \approx 0$. **Horn: no amplitude / no steering** (C3S.json, 324f097). - **S-F3** (local bias $\times$ chord kernel, $|\cdot|$ ansatz with κ): steers ($D > 0$) but the marginal carries Bob-setting dependence at $O(\kappa\chi)$ — telegraph 0.500 $\rightarrow$ 0.542 (C3S_F3_check.py; analytics 2053106). **Horn: superluminal telegraph** — kin to finite-speed no-gos [BPA12].

Reduction (Bell corollary, not a new theorem). The internal class = λ -average of local responses = LHV $\Rightarrow S \leq 2$. Closing the seam therefore demands **joint, nonlocal resolution**; and joint resolution demands a shared, statistically invisible precedence structure (§6). Attempt count: 3. (C3S_seam_reduction.md.)

6. FRAME — foliation (C3-L, L1/L2)

On the chord joint law, two resolution mechanics: - **(i) preferred-foliation:** $S = 2\sqrt{2}$ **identical across five foliations** (cross-disc 0.000000 under common random numbers), foliation statistically invisible (C3L_L2.json, 49e8c1b). - **(ii) frame-local:** space-like $\rightarrow$ LHV ceiling $S = 0.943$; time-like $A \rightarrow B$, $B \rightarrow A \rightarrow$ reconstruct $2\sqrt{2}$. L1: the joint admits an order-invariant realization (existential PASS, factorized witness disc = 0.00); the collapse realization's order-dependence (0.0418 with invariant marginals) is evidence for L2c, not pathology (addendum1). **Method note:** the foliation-invariance test required common random numbers — a raw cross-range would false-STOP on Monte-Carlo noise (§7).

Class-M theorem (status: theorem; canonical statement in sim/cycle3/C3L_L2c_THEOREM.md, c8e1bf3): in the class M of outcome-definite sequential resolution mechanisms (shared preparation randomness λ , setting-independence M4', no retrocausation), $S > 2$ is achievable **iff** M admits a shared, space-like-transcending, statistically invisible precedence structure. Realizations demonstrated: a preferred foliation (five axes, cross-discrimination 0.000000 under CRN) and a **frameless per-run shared order-coin** ($S = 2\sqrt{2} \pm 2\sigma$, no telegraph, coin invisible; battery c8e1bf3,

independently re-executed). The earlier conjecture’s “preferred foliation is required” was **retracted as too strong** during the adversarial passes: the foliation is one realization of the required structure, not its only form. Experimental anchor unchanged: before-before experiments [ZBGT01, SZGS02] exclude the frame-local branch in nature; what survives is joint resolution carrying a shared, hidden precedence structure. Kin: Bohmian preferred slicing [DGZ92], steering-axiomatic reconstructions [CDP11]; [H92], [SS97].

7. METHODS — batteries, controls, self-correction

All batteries: pure numpy, $N=2 \cdot 10^6$, PRNGKey from config, analytics beside numerics ($>2\sigma$ mismatch = stop), assert prereg $\leftrightarrow$ JSON before raw, raw $\rightarrow$ commit $\rightarrow$ analysis. Estimator named in every prediction; metric = the model observable.

Registered false stops and their resolution (part of the evidence): 1. **$\Delta(2)>0$ = bug** (C3-B): normalized conditional axis broke total probability; fixed to raw Bloch (968aaa5). 2. **Quadrature gate** (hardening): an analytic-tolerance gate (tol=1e-9 on a 20001-pt trapezoid, error $\sim 2.3e-5$) false-STOPped T1/T2; corrected to the registered $<2\sigma$ criterion + algebraic zero at $p=2$; threshold **not** loosened to pass (addendum2, ccf5af6). 3. **Monte-Carlo cross-range** (L2): raw 5-foliation range would false-STOP; replaced by common-random-number comparison (49e8c1b). 4. **§2.3 over-claim retraction** (C3-S): the first §2.3 asserted “no-signaling alone cuts $p>2$ in the joint layer”; J5 rediscovered the Popescu-Rohrlich fact that the postulated $|\cdot|^p$ family is no-signaling up to the PR box [PR94]; the claim was narrowed to steering-endowed layers (324f097; §8). 5. **Frame over-claim** (abstract/§5/§8/§9 v1): “preferred frame required” — retracted by the class-M theorem’s T3’ and the β -coin demonstration; revised in v2, history retained. §9 CEILING wording aligned in v2. Each was caught by a preregistered criterion, not by hindsight.

Reproduction. The full numpy battery suite of cycle 3 was independently re-executed (2026-07-21); fixed-seed raw reproduces bitwise (C3-B B1–B7, C3-B hardening, L3, L1, L2, C3-S, β -coin — all results JSON byte-identical under `git diff`). C3-B and the β -coin were additionally reproduced by the architect in a separate environment.

8. The Tsirelson section (§2.3, corrected — verbatim)

Rule selection is exactly as strong as steering. In any layer whose measurements produce conditional states that can themselves be steered — internally generated diversity $D>0$ — the Jensen argument applies: no-signaling forces affinity, selects $p=2$, and with it the quantum value of S available to that layer. In a bare-correlation joint layer, however, where outcomes carry no post-measurement structure, the $|\cdot|^p$ family has uniform marginals for every p : the entire family is no-signaling, up to and including the PR box — the classic observation of Popescu and Rohrlich [PR94], which our battery rediscovered as a registered finding against an earlier over-claim of this section (retained in the record). No-signaling alone therefore does not enforce the Tsirelson bound; what enforces it, in this geometry, is no-signaling PLUS steering structure — and this localizes, within one construction, the same gap that information causality [Paw09] was invented to fill. The amplitude seam then resolves into a

reduction rather than a mystery: within the internally-generated class (hidden measure plus local response) Bell's theorem itself caps S at 2, so closing the seam — $D > 0$ together with $S = 2\sqrt{2}$ — demands joint, non-local resolution of the pair; and joint resolution, by the foliation results above, demands a preferred frame that the statistics provably hide. [Revised by the class-M theorem, §6: read 'a shared, statistically invisible precedence structure', of which a preferred frame is one realization — the frameless order-coin is another.] The program's bridge statement follows: Einstein's prohibition selects the quantum rule wherever mechanisms carry steering; the quantum amplitude is bought only by joint resolution, whose frame Einstein's own statistics render invisible.

9. Outlook (verbatim)

Three cycles of this program asked one question from three sides: what stands between a mechanism and quantum correlations. The answer assembled itself into four registered statements. (1) RULE: wherever a mechanism carries steering — conditional states with internally generated diversity — the prohibition of superluminal signaling forces the Born exponent $p=2$ uniquely; the derivation is internal, executable, and robust to deformations of the source measure. (2) CEILING: mechanisms that resolve outcomes locally (hidden measure plus local response) are capped by Bell's theorem at $S=2$, with the $1/3$ -cosine point as their isotropic-cosine ceiling — a landmark anchored analytically (the lever rule), reached by frame-local resolution, and bracketed by relaxational dynamics (estimator-dependent, §4). (3) SEAM: the gap between that ceiling and the quantum value is not crossed by decorating local mechanisms — our registered attempts fail on a trilemma of horns (no steering, no amplitude, or superluminal telegraph; kin to finite-speed no-gos [BPA12]) — it is crossed only by joint resolution of the pair. (4) FRAME: joint resolution of the frame-local kind reconstructs the Suarez-Scarani alternative and degrades to the local ceiling, which real before-before experiments [ZBGT01, SZGS02] have excluded in nature; the surviving corner, joint resolution carrying a shared, statistically invisible precedence structure (foliation or shared order-coin), reproduces quantum statistics exactly while rendering its own precedence structure statistically invisible to the resolution of our batteries. The bridge statement of the program is therefore not a unification but an exact division of labor: Einstein's causality selects the quantum rule; the quantum amplitude is bought only by joint resolution; and the precedence structure that joint resolution requires is precisely what the selected statistics forever hide. What remains open is stated as precisely: a mechanical realization of the asymmetry parameter (C3-B-mech), the structural theorem behind the seam trilemma (the class-M theorem, proved; N-event generalization = C4 G-T), and the ground-state problem of the stiff chain — walls mapped, kill criteria attached, commits public.

Appendix A — batteries (reproducible)

Battery	Prereg	Raw	Result
C3-B B1–B7	dd7a74e	968aaa5	steering generated, $p=2$ forced
C3-B hardening T1–T3	96a3178	c152c98	internal derivation ($T1\checkmark T2\checkmark$)
L3 scan + bridge	2d78c51	9567ddb	tail $\Delta>0$; internal $S(2)=\sqrt{2}$
C3-L L1/L2	0b10f5d,100a261	1e0cf0a,49e8c1b	foliation invisible; frame- local→ceiling
C3-S J1–J7	3d516d4	324f097	seam not closed (S-F1/S-F2)
S-F3 analytics	—	2053106	closed pre-battery (telegraph)
β -coin (class M)	C3L_coin_prereg	c8e1bf3	frameless $2\sqrt{2}$, invisible
Class-M theorem	—	c8e1bf3	achievability iff shared invisible precedence
C3-B-mech M0 (calibration)	C3Bmech addenda	0a6c51e	source polarizes; $\chi(h)$ linear
C3-B-mech M1/M2	63456cd	2080bbe	mechanical steering, form-free ($M1\checkmark M2\checkmark$)

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